Day 11 — Splunk Fundamentals (LetsDefend)

📅 Date

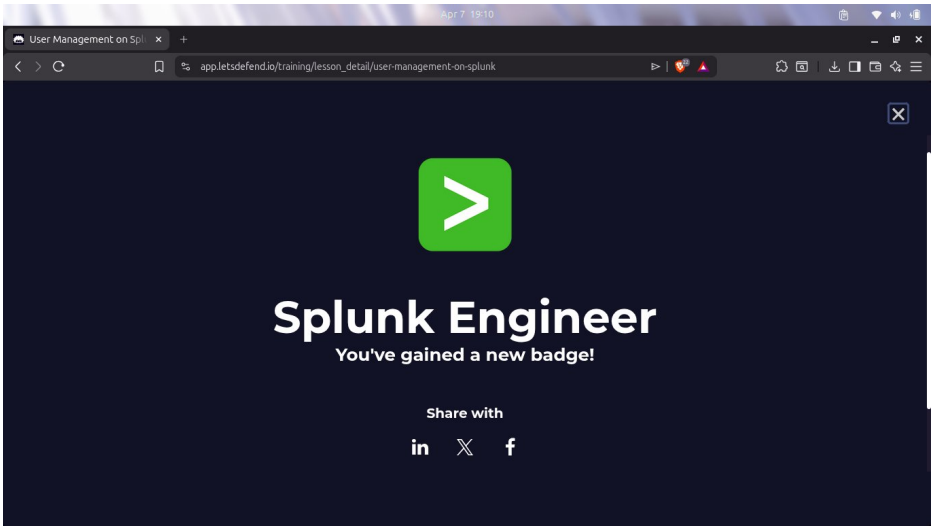
April 7, 2026

🎯 Platform

- LetsDefend.io (Free Tier)
- Course: Introduction to Splunk

🏆 Badge Earned

- **Splunk Engineer** — LetsDefend

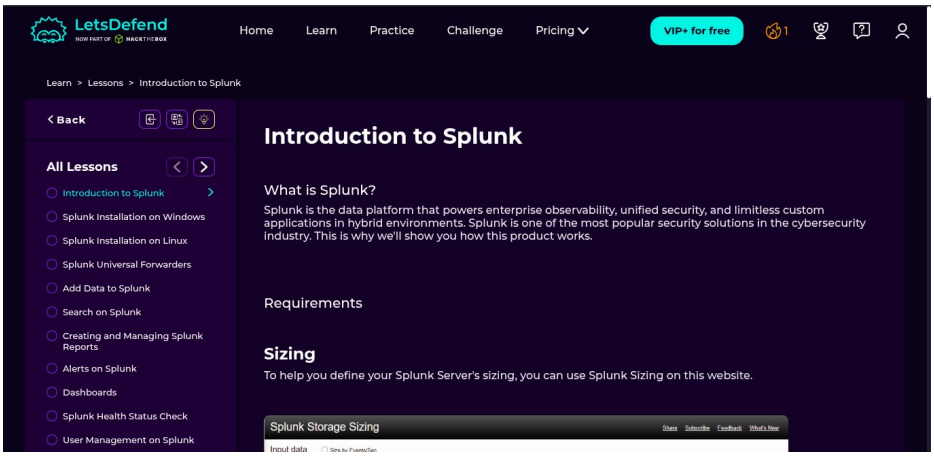


Splunk Engineer Badge

📚 Topics Covered

#	Lesson	Status
1	Introduction to Splunk	✅ Completed
2	Splunk Installation on Windows	✅ Completed
3	Splunk Installation on Linux	✅ Completed
4	Splunk Universal Forwarders	✅ Completed
5	Add Data to Splunk	✅ Completed
6	Search on Splunk	✅ Completed
7	Creating and Managing Splunk Reports	✅ Completed
8	Alerts on Splunk	✅ Completed
9	Dashboards	✅ Completed
10	Splunk Health Status Check	✅ Completed
11	User Management on Splunk	✅ Completed

🔍 What is Splunk?



Intro to Splunk

Splunk is a data platform that powers enterprise observability, unified security, and custom applications in hybrid environments. It is one of the most widely used SIEM solutions in the cybersecurity industry.

Key Components

- **Indexer** — Stores and indexes ingested log data
- **Search Head** — Interface for running SPL queries and investigations

- **Universal Forwarder** — Lightweight agent that collects and forwards logs to the indexer
- **SPL** — Splunk Processing Language, used to search and analyze data

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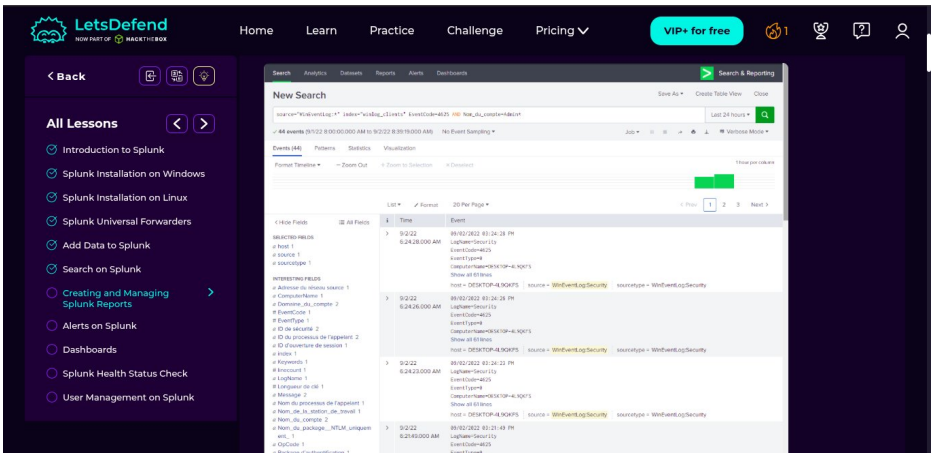
🔎 Search on Splunk



Search on Splunk

Splunk's Search & Reporting interface allows analysts to query indexed data using SPL.

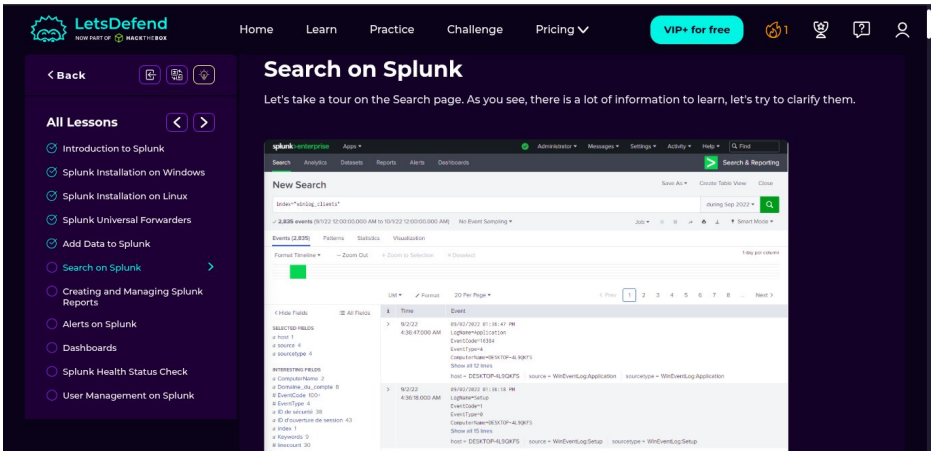
Key SPL Concepts Learned

```
-- Search all events in an index
index="winlog_clients"

-- Filter by EventCode and username
source="WinEventLog:*" index="winlog_clients" EventCode=4625 AND
Nom_du_compte=AdminT

-- View events in a time range
index="winlog_clients" earliest=-24h
```

Key Fields Observed

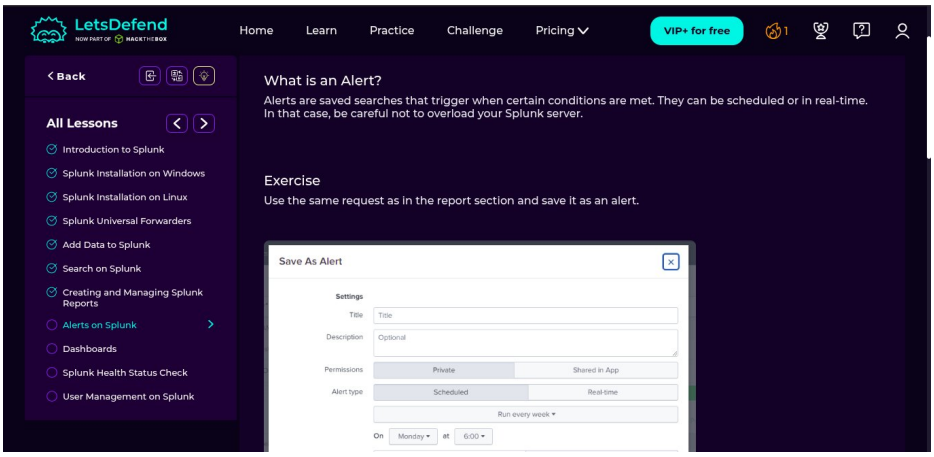
- `host` — The machine that generated the log
- `source` — Log source (e.g. WinEventLog:Security)
- `sourcetype` — Type of log
- `EventCode` — Windows Event ID
- `ComputerName` — Name of the machine

📊 SPL Query in Action



SPL Query

🔔 Alerts on Splunk



Alerts on Splunk

- Alerts are saved searches that **automatically trigger** when conditions are met
- Can be scheduled or run in real-time
- Alert types: **Scheduled** or **Real-time**
- Care must be taken not to overload the Splunk server with too many real-time alerts

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📈 Dashboards

- Dashboards visualize SPL query results
- Useful for SOC monitoring and presenting data to management
- Can include charts, tables, and single value panels

👥 User Management

- Splunk supports role-based access control (RBAC)
- Roles include: Admin, Power, User
- User permissions control what data and features each analyst can access

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🔄 Wazuh vs Splunk

Feature	Wazuh	Splunk
Cost	Free & Open Source	Free tier limited
Log Ingestion	Agent-based	Agent + Forwarder
Search Language	Lucene-based	SPL
Alerting	Built-in rules	Saved searches
Dashboards	Kibana	Splunk Web
Best For	Small labs, open source	Enterprise SOC

🏷️ MITRE ATT&CK Mapping

Technique	ID	Relevance
Valid Accounts	T1078	Detected via EventCode 4625 failed logins
Brute Force	T1110	Multiple failed login alerts

💡 Key Takeaways

1. Splunk is one of the most in-demand SIEM tools in enterprise SOC environments
2. SPL is powerful for filtering, correlating, and visualizing log data
3. Universal Forwarders make it easy to collect logs from multiple sources
4. Alerts and dashboards automate detection and improve analyst efficiency
5. Role-based access control is critical in a real SOC environment

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🔗 Resources

- LetsDefend Splunk Course
- Splunk SPL Documentation
- MITRE ATT&CK